Stability surface for confinated hydrogel layer on substrates

Duvan Henao* Pedro Hernández-Llanos†‡ Carlos Román§
Xavier Lamy¶

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Abstract

This paper combines rigorous mathematical analysis with finite element simulations and computational modelling of nonlinear gel systems. We study a generalization of the models considered by Duvan Henao et al. in [1, 2, 3, 4, 12], which incorporate the chemical potential $\hat{\mu}$ and a single material parameter ϕ_0, namely, the polymer volume fraction in the reference configuration. Our model also generalizes the chemical-potential formulation studied by Kang & Huang in JMPS 2010 [9], where instability was reported. In contrast, our analysis and numerical simulations show that the undulation instabilities reported in the literature depend strongly on the polymer reference volume fraction ϕ_0: the model becomes stable for sufficiently low values of ϕ_0. This result is obtained through finite element simulations and Fourier decomposition of the linearized equations around the homogeneous uniaxial extension.

The stability result constitutes a major step towards the desired Γ-convergence justification of the asymptotic debonding energy-release formula previously obtained by Duvan Henao, Calderer, Sánchez, Siegel, and Song (J. Elast. 2020 [1]; Polymer 2025 [4]).

Key words: Gels, Debonding, Thin film, Nonlinear elasticity, Flory-Huggins

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*Instituto de Ciencias de la Ingeniería, Universidad de O'Higgins, Rancagua, Chile. E-mail: duvan.henao@uoh.cl

†Instituto de Ciencias de la Ingeniería, Universidad de O'Higgins, Rancagua, Chile. E-mail: pedro.hernandez@postdoc.uoh.cl

‡Department of Mathematics and Mathematical Statistics, Umeå University, Umeå, Sweden. E-mail: phernandezllanos@gmail.com

§Departamento de Matemáticas, Pontificia Universidad Católica de Chile, Santiago, Chile. Email: carlos.roman@mat.uc.cl

¶Institut de Mathématiques de Toulouse, Université de Toulouse, France. Email: xlamy@math.univ-toulouse.fr

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1 A nonlinear theory for hydrogels

Regardless of the transport kinetics, deformation of the hydrogel eventually reaches an equilibrium state when both the chemical potential and the mechanical stress satisfy the equilibrium condition. The chemical equilibrium requires that the chemical potential inside the gel be a constant and equal to the chemical potential of the external solvent ($\mu = \hat{\mu}$. In addition the mechanical equilibrium requires that

$$\frac{\partial S_{iJ}}{\partial X_J} + B_i = 0 \text{ in } \Omega_0, \tag{1.1}$$

and

$$S_{iJ} N_J = T_i \text{ or } \delta x_i = 0 \text{ on } \Gamma, \tag{1.2}$$

where SiJ is the nominal stress, B_i and T_i the nominal body force and surface traction, Ω_0 and Γ the volume and the surface of the hydrogel at the reference state, N_J the unit vector in the outward normal direction of the surface,X_J and x_i the particles coordinates at the reference and the current states, respectively, and they are related to each other through the deformation gradien tensor $F_{iJ} = dx_i/dX_J$. Here the reference state is taken to be the undeformed dry state of the polymer network, and the body force is considered negligible.

The chemical potential of the external solvent ($\hat{\mu}$) in general depends on the temperature (T) and pressure (p). Assuming an ideal gas phase ($p < p_0$) and an incompressible liquid phase ($p > p_0$) for the solvent, the chemical potential is given by

$$\hat{\mu}(p,T) = \begin{cases} (p - p_0)\nu, & \text{if } p > p_0; \\ K_B T \ln(p/p_0), & \text{if } p < p_0, \end{cases} \tag{1.3}$$

where p_0 is the equilibrium vapor pressure and depends on the temperature, ν the volume per solvent molecule, and K_B the Boltzmann constant. At the equilibrium vapor pressure ($p = p_0$), the external chemical potential $\hat{\mu} = 0$. For water at $25°C$, $p_0 \sim 3.2KPa$ and $\nu \sim 3{\times}10^{-29}m^3$, the chemical potential is about $0.00007K_BT$ under the atmospheric pressure ($\sim 101KPa$). In a vacumm $(p = 0), \hat{\mu} = -\infty$.

The constitutive behavior of the hydrogel can be described by using a free-energy function U, which in general depends on both the elastic deformation of the polymer network an the concentration of solvent molecules inside the gel. The nominal stress in the gel is

$$S_{iJ} = \frac{\partial U}{\partial F_{iJ}}. \tag{1.4}$$

To be specific, we adopt a free-energy density function that consists of two separate parts; one for elastic deformation of polymer network and the other for mixing of solvent molecules with the polymer, namely;

$$U(\mathbf{F}, C) = U_e(\mathbf{F}) + U_m(C). \tag{1.5}$$

$$U_e(\mathbf{F}) = \frac{1}{2} N K_B T (I - 3 - 2\ln J), \tag{1.6}$$

$$U_m(C) = \frac{K_B T}{\nu} \left(\nu C \ln \frac{\nu C}{\phi_0 + \nu C} + \frac{\chi \nu C \phi_0}{\phi_0 + \nu C} \right), \tag{1.7}$$

and $I = F_{iJ}F_{iJ}$, $J = \det(\mathbf{F})$, C is the nominal concentration (number per volume) of solvent molecules, N the effective number of polymer chain per unit volume of the hydrogel at the dry state, and χ a dimensionless quantity characterizing the interaction energy between the solvent molecules and the polymer. Therefore, the material properties of the hydrogel system are fully determinated by the three parameters: $NK_BT, K_BT/\nu$ and χ, the first of which is simply the initial shear modulus of the polymer network.

The elastic free-energy density function (U_e) in Eq.(1.6) is similar to that obtained by Flory (1953) based on a statistical mechanics model, but is slightly different as suggested in the previous works (Hong et al.,2008; Kang and Huang, 2010). The free energy of mixing (U_m) in Eq.(1.7) is derived exactly from the Flory-Huggins polymer solution theory (Huggins, 1941; Flory, 1953), with the assumption of molecular incompressibility inside the gel, namely, the total volume of the hydrogel is the sum of the volume of the dry polymer network and the solvent molecules so that

$$J = \phi_0 + \nu C. \tag{1.8}$$

At the equilibrium state of swelling, the chemical potential inside of the hydrogel is a constant while the concentration field can be inhomogeneous. Thus it is a convenient to write the free-energy function in terms of the chemical potential via a Legendre transformation (Hong et al. 2009a), namely

$$\hat{U}(\mathbf{U}, C) = U(\mathbf{U}, C) - \hat{\mu} C. \tag{1.9}$$

In then follow by Eqs.(1.11)-(1.8)

$$S_{iJ} = \frac{\partial \hat{U}}{\partial F_{iJ}} = N K_B T (F_{iJ} + \alpha H_{iJ}), \tag{1.10}$$

where

$$\alpha = -\frac{1}{J} + \frac{1}{N\nu}\left(\ln\frac{J-\phi_0}{J} + \frac{\phi_0}{J} + \frac{\chi\phi_0^2}{J^2} - \frac{\hat{\mu}}{K_B T}\right), \tag{1.11}$$

$$H_{iJ} = \frac{\partial J}{\partial F_{iJ}} = \frac{1}{2} e_{ijk} e_{JKL} F_{jK} F_{kL}. \tag{1.12}$$

2 Homogeneous swelling of a confined hydrogel layer

The volume swelling ratio is simply, $J_h = \lambda_h$, and the nominal concentration of solvent molecule in the hydrogel is

$$C_h = \frac{\lambda_h - \phi_0}{\nu} \tag{2.1}$$

Upon swelling, the free-energy density inside the hydrogel becomes

$$U_h(\lambda_h) = U_e(\lambda_h) + U_m(C_h), \tag{2.2}$$

where, by

$$U_e(\lambda_h) = \frac{1}{2} N K_B T[\lambda_h^2 - 1 - 2\ln(\lambda_h)], \tag{2.3}$$

$$U_m(C_h) = \frac{K_B T}{\nu}\left[(\lambda_h - \phi_0)\ln\left(1 - \frac{\phi_0}{\lambda_h}\right) + \chi\phi_0\left(1 - \frac{\phi_0}{\lambda_h}\right)\right]. \tag{2.4}$$

The total free energy of the system (including the hydrogel and the external solvent) consist of the internal free energy and the chemical/mechanical work done during absorption an swelling, namely

$$G = U_h V_0 - \hat{\mu} C_h V_0 + (J_h - \phi_0) p V_0 = \left[U_h(\lambda_h) - \frac{\hat{\mu} - p\nu}{\nu}(\lambda_h - \phi_0)\right] V_0. \tag{2.5}$$

where V_0 is the reference volume of the layer at the dry state. Note that the external solvents exerts a pressure p onto the surface hydrogel layer, which does a negative work and thus increases the free energy as the hydrogel swells. The equilibrium swelling ratio of the hydrogel can then be determinated by minimizing the total free energy. Setting $dG/d\lambda_h = 0$ leads to

$$\ln\left(1 - \frac{\phi_0}{\lambda_h}\right) + \frac{\phi_0}{\lambda_h} + \frac{\chi\phi_0^2}{\lambda_h^2} + N\nu\left(\lambda_h - \frac{1}{\lambda_h}\right) = \frac{\hat{\mu} - p\nu}{K_B T} \tag{2.6}$$

Solving Eq.(2.6) gives the homogeneous swelling ratio λ_h for a specific hydrogel system as a function of the external chemical potencial ($\hat{\mu}$), which in turns is a function of the temperature and pressure. By the definition of the chemical potential in Eq., we note that, if the external solvent is in a liquid phase (i.e. $p > p_0$), the righ-hand side of Eq.(2.6) is independent of the pressure p and thus the homogeneous swelling ratio depends on the temperature only, while this is not the case if $p < p_0$ (a gas phase). In the previous studies (Hong et.al. 2009a; Kang and Huang, 2010), a similar solution was obtained by using a Lagrange multiplier, but the effect of the external pressure was ignorated. Fig 3 plots the homogeneous swelling ratio as a function of the chemical potential, comparing the present solution to the previous solution. In addition to the two dimensionless materials parameters ($N\nu$ and χ) for the hydrogel, the present solution depends on the normalized equilibrium vapor pressure, $\overline{p_0} = p_0\nu/(K_BT)$, which is a function of the temperature for a specific solvent. At a constant temperature, the hidrogel swells incresingly as the chemical potential increses until the pressure reaches the equilibrium vapor pressure ($p = p_0$ and $\mu = 0$). For a typical value of the equilibrium vapor pressure (e.g. for water at $25°C$, $\overline{p_0} \sim 2.3 \times 10^{-5}$), the difference between the two solutions is negligible when $\mu \leq 0$ or $p \leq p_0$. However, when $p > p_0$, the previous solution predicts that the swelling ratio continues to increase as the chemical potential or pressure increases, while the present solution predicts a constan swelling ratio independent of the chemical potential or pressure. Consequently, given the material parameters for a hydrogel system ($N\nu$ and χ), the maximum degree of homogeneous swelling occurs at the equilibrium vapor pressure ($\mu = 0$). Also plotted in Fig 3. is the homogeneous swelling ratio for an artificially large equilibrium vapor pressure ($\overline{p_0} = 0.01$), to ilustrate the effect of solvent vapor pressure on the swelling ratio, i.e., λ_h decreases as $\overline{p_0}$ increases. Similar effect can be predicted for swelling of unconstrained hydrogels. The lateral confinements by the substrate induces a compressive stress in the swollen hydrogel layer. By Eqs.(1.12) and (2.6), we obtain that

$$S_{iJ} = NK_BT\left(F_{iJ} - \lambda_h H_{iJ}\right) - pH_{iJ}. \tag{2.7}$$

3 Linear perturbation analysis

3.0.1 Choice of deformations

We consider deformations $y = (y_1, y_2, y_3)$ given by

$$\begin{aligned}
y_1(\xi_1,\xi_2,\xi_3) &= \xi_1 + u_1(\xi_1, \lambda_h\xi_2),\\
y_2(\xi_1,\xi_2,\xi_3) &= \lambda_h\xi_2 + u_2(\xi_1, \lambda_h\xi_2),\\
y_2(\xi_1,\xi_2,\xi_3) &= \xi_3.
\end{aligned}$$

$$\widetilde{F} = \frac{\partial(y_1, y_2, y_3)}{\partial(\xi_1, \xi_2, \xi_3)} = \nabla y = \begin{pmatrix} 1 + \dfrac{\partial u_1}{\partial x_1} & \lambda_h \dfrac{\partial u_1}{\partial x_2} & 0 \\ \dfrac{\partial u_2}{\partial x_1} & \lambda_h \left(1 + \dfrac{\partial u_2}{\partial x_2}\right) & 0 \\ 0 & 0 & 1 \end{pmatrix}$$

Then,

$$\widetilde{J} = \det \widetilde{F} \approx \lambda_h \left(1 + \frac{\partial u_1}{\partial x_1} + \frac{\partial u_2}{\partial x_2}\right) = \lambda_h(1 + \varepsilon). \tag{3.1}$$

If $\widetilde{J} \approx \lambda_h$, then

$$\alpha(\widetilde{J}) \approx \alpha(\lambda_h) + \alpha'(\lambda_h)(\widetilde{J} - \lambda_h) = \alpha(\lambda_h) + \lambda_h \alpha'(\lambda_h)\varepsilon.$$

Since $\alpha(\lambda_h) = -\lambda_h$ and α as in (1.11). We can define $\varphi = \dfrac{\phi_0}{\widetilde{J}}$, then we have

$$\alpha = -\frac{\varphi}{\phi_0} + \frac{1}{\gamma}\left[\ln(1-\varphi) + \varphi + \chi\varphi^2 + \frac{V_m}{K_B T} p_0\right],$$

where $\varphi = \phi_0/J$, then

$$\frac{d\alpha}{d\varphi} = -\frac{1}{\phi_0} + \frac{\varphi}{NV_m}\left(2\chi - \frac{1}{1-\varphi}\right).$$

Therefore,

$$\alpha'(\lambda_h) = \frac{d\alpha}{d\varphi} \cdot \frac{d\varphi}{dJ}\bigg|_{J=\lambda_h}. \tag{3.2}$$

Regarding (3.1) then we have by virtue of (3.2) that

$$\begin{aligned}
\alpha(\widetilde{J}) &\approx \alpha(\lambda_h) + \lambda_h \alpha'(\lambda_h)\varepsilon = -\lambda_h + \lambda_h \varepsilon \frac{d\alpha}{d\varphi} \cdot \frac{d\varphi}{dJ}\bigg|_{J=\lambda_h} \\
&= -\lambda_h + \lambda_h \varepsilon \left[-\frac{1}{\phi_0} + \frac{\varphi}{NV_m}\left(2\chi - \frac{1}{1-\varphi}\right)\right] \cdot \left(-J^{-2} \cdot \phi_0\right)\big|_{J=\lambda_h} \\
&= -\lambda_h - \varepsilon \left(-\frac{1}{\lambda_h} + \frac{\phi_0^2}{\lambda_h^2 NV_m}\left(2\chi - \frac{1}{1 - \dfrac{\phi_0}{\lambda_h}}\right)\right) \\
&= -\lambda_h + \varepsilon \frac{1}{\lambda_h} - \frac{\varepsilon \phi_0^2}{\lambda_h NV_m}\left(\frac{2\chi}{\lambda_h} - \frac{1}{\lambda_h - \phi_0}\right) \\
&= -\lambda_h + \varepsilon \left[\frac{1}{\lambda_h} + \frac{\phi_0^2}{\gamma}\left(\frac{1}{\lambda_h(\lambda_h - \phi_0)} - \frac{2\chi}{\lambda_h^2}\right)\right].
\end{aligned} \tag{3.3}$$

In the right-hand expression of (3.3) we can apply partial fractions as follows

$$\alpha(\widetilde{J}) \approx -\lambda_h + \varepsilon\frac{1}{\lambda_h} + \varepsilon\frac{\phi_0^2}{\gamma}\left(\frac{A}{\lambda_h} + \frac{B}{(\lambda_h - \phi_0)} - \frac{2\chi}{\lambda_h^2}\right),$$

Where do we get that

$$A\lambda_h + B\lambda_h - B\phi_0 = 1 \to (A+B)\lambda_h - B\phi_0 = 1,$$

now $A + B = 0$ and $-B\phi_0 = 1$. Therefore, $A = -\dfrac{1}{\phi_0}$ and $A = \dfrac{1}{\phi_0}$.

Finally, we obtain

$$\alpha(\widetilde{J}) \approx -\lambda_h + \varepsilon\left[\frac{1}{\lambda_h} + \frac{\phi_0^2}{\gamma}\left(\frac{1}{\phi_0(\lambda_h - \phi_0)} - \frac{1}{\phi_0\lambda_h} - \frac{2\chi}{\lambda_h^2}\right)\right].$$

Let ξ_h be defined as follows

$$\xi_h = \frac{1}{\lambda_h} + \frac{\phi_0^2}{\gamma}\left(\frac{1}{\phi_0(\lambda_h - \phi_0)} - \frac{1}{\phi_0\lambda_h} - \frac{2\chi}{\lambda_h^2}\right), \tag{3.4}$$

when $\phi_0 = 1$ we have the Eq.(4.6) in [9].

Therefore,

$$P \approx G\left[\widetilde{F} + \alpha(\widetilde{J})(\mathrm{Cof}\,\widetilde{F})\right] = G\left[\widetilde{F} - (\lambda_h - \xi_h\varepsilon)(\mathrm{Cof}\,\widetilde{F})\right].$$

Now, we are interested in computing the components of P in order to proced to the analysis

$$\begin{aligned}
P_{11} &\approx G\left[\widetilde{F}_{11} - (\lambda_h - \xi_h\varepsilon)(\mathrm{Cof}\,\widetilde{F})_{11}\right],\\
&= G\left[1 + \frac{\partial u_1}{\partial x_1} - \left(\lambda_h - \xi_h\left(\frac{\partial u_1}{\partial x_1} + \frac{\partial u_2}{\partial x_2}\right)\right)\lambda_h\left(1 + \frac{\partial u_2}{\partial x_2}\right)\right]\\
&= G\left[1 - \lambda_h^2 - \lambda_h^2\frac{\partial u_2}{\partial x_2} + \lambda_h\xi_h\frac{\partial u_2}{\partial x_2} + \lambda_h\xi_h\frac{\partial u_1}{\partial x_1} + \frac{\partial u_1}{\partial x_1} + \text{h.o.t.}\right],
\end{aligned}$$

therefore,

$$P_{11} \approx P_h + G\left[(1 + \lambda_h\xi_h)\frac{\partial u_1}{\partial x_1} + \lambda_h(\xi_h - \lambda_h)\frac{\partial u_2}{\partial x_2}\right]. \tag{3.5}$$

$$\begin{aligned}
P_{22} &\approx G\left[\widetilde{F}_{22} - (\lambda_h - \xi_h\varepsilon)(\mathrm{Cof}\,\widetilde{F})_{22}\right],\\
&= G\left[\left(\lambda_h + \lambda_h\frac{\partial u_2}{\partial x_2}\right) - \left(\lambda_h - \xi_h\left(\frac{\partial u_1}{\partial x_1} + \frac{\partial u_2}{\partial x_2}\right)\right)\left(1 + \frac{\partial u_1}{\partial x_1}\right)\right]\\
&= G\left[\lambda_h + \lambda_h\frac{\partial u_2}{\partial x_2} - \lambda_h - \lambda_h\frac{\partial u_1}{\partial x_1} + \xi_h\frac{\partial u_1}{\partial x_1} + \xi_h\frac{\partial u_2}{\partial x_2} + \text{h.o.t.}\right],
\end{aligned}$$

therefore,

$$P_{22} \approx G\left[(\xi_h - \lambda_h)\frac{\partial u_1}{\partial x_1} + (\xi_h + \lambda_h)\frac{\partial u_2}{\partial x_2}\right]. \tag{3.6}$$

$$\begin{aligned}
P_{33} &\approx G\left[\widetilde{F}_{33} - (\lambda_h - \xi_h\varepsilon)(\operatorname{Cof}\widetilde{F})_{33}\right], \\
&= G\left[1 - (\lambda_h - \xi_h\varepsilon)(\lambda_h + \lambda_h\varepsilon)\right] \\
&= G\left[1 - \lambda_h^2 - \lambda_h^2\varepsilon + \xi_h\lambda_h\varepsilon + \lambda_h\xi_h\varepsilon^2\right], \\
&= -G\left(\lambda_h^2 - 1\right) + G\left[(\xi_h - \lambda_h)\varepsilon\lambda_h\right] + \text{h.o.t.}
\end{aligned}$$

therefore,

$$P_{33} \approx P_h + G\left[(\xi_h - \lambda_h)\varepsilon\lambda_h\right]. \tag{3.7}$$

$$\begin{aligned}
P_{12} &\approx G\left[\widetilde{F}_{12} - (\lambda_h - \xi_h\varepsilon)(\operatorname{Cof}\widetilde{F})_{12}\right], \\
&= G\left[\lambda_h\frac{\partial u_1}{\partial x_2} - (\lambda_h - \xi_h\varepsilon)\left(-\frac{\partial u_2}{\partial x_1}\right)\right] \\
&= G\left[\lambda_h\frac{\partial u_1}{\partial x_2} + \lambda_h\frac{\partial u_2}{\partial x_1} - \xi_h\varepsilon\frac{\partial u_2}{\partial x_1}\right] \\
&= G\lambda_h\left[\frac{\partial u_1}{\partial x_2} + \frac{\partial u_2}{\partial x_1}\right] + \text{h.o.t.}
\end{aligned}$$

therefore,

$$P_{12} \approx G\lambda_h\left[\frac{\partial u_1}{\partial x_2} + \frac{\partial u_2}{\partial x_1}\right]. \tag{3.8}$$

$$\begin{aligned}
P_{21} &\approx G\left[\widetilde{F}_{21} - (\lambda_h - \xi_h\varepsilon)(\operatorname{Cof}\widetilde{F})_{21}\right], \\
&= G\left[\frac{\partial u_2}{\partial x_1} - (\lambda_h - \xi_h\varepsilon)\left(-\lambda_h\frac{\partial u_1}{\partial x_2}\right)\right] \\
&= G\left[\frac{\partial u_2}{\partial x_1} + \lambda_h^2\frac{\partial u_1}{\partial x_2} - \lambda_h\xi_h\varepsilon\frac{\partial u_1}{\partial x_2}\right] \\
&= G\left[\lambda_h^2\frac{\partial u_1}{\partial x_2} + \frac{\partial u_2}{\partial x_1}\right] + \text{h.o.t.}
\end{aligned}$$

therefore,

$$P_{21} \approx G\left[\lambda_h^2\frac{\partial u_1}{\partial x_2} + \frac{\partial u_2}{\partial x_1}\right]. \tag{3.9}$$

Analogously, we can obtain

$$P_{23} = P_{32} = P_{13} = P_{31} = 0, \tag{3.10}$$

since $\widetilde{F}_{\alpha 3} = \widetilde{F}_{3\alpha} = 0$ and $(\mathrm{Cof}\,\widetilde{F})_{\alpha 3} = (\mathrm{Cof}\,\widetilde{F})_{3\alpha} = 0$ for $\alpha \in \{1, 2\}$.

In the other hand we consider the following equilibrium condition

$$\frac{\partial P_{ij}}{\partial X_j} = 0. \tag{3.11}$$

By application of (3.11) to (3.5)-(3.8) we have

$$0 = \frac{\partial P_{11}}{\partial X_1} \approx \cancelto{0}{\frac{\partial P_h}{\partial x_1}} + G\left[(1+\lambda_h\xi_h)\frac{\partial^2 u_1}{\partial x_1^2} + \lambda_h(\xi_h - \lambda_h)\frac{\partial^2 u_2}{\partial x_1 \partial x_2}\right], \tag{3.12}$$

$$0 = \frac{\partial P_{22}}{\partial X_2} \approx G\lambda_h\left[(\xi_h - \lambda_h)\frac{\partial^2 u_1}{\partial x_2 \partial x_1} + (\xi_h + \lambda_h)\frac{\partial^2 u_2}{\partial x_2^2}\right], \tag{3.13}$$

$$0 = \frac{\partial P_{21}}{\partial X_1} \approx G\left[\lambda_h^2\frac{\partial^2 u_1}{\partial x_1 \partial x_2} + \frac{\partial^2 u_2}{\partial x_1^2}\right], \tag{3.14}$$

$$0 = \frac{\partial P_{12}}{\partial X_2} \approx G\lambda_h^2\left[\frac{\partial^2 u_1}{\partial x_2^2} + \frac{\partial^2 u_2}{\partial x_2 \partial x_1}\right]. \tag{3.15}$$

Finally, combining equation (3.12) and (3.15) we obtain

$$(1+\lambda_h\xi_h)\frac{\partial^2 u_1}{\partial x_1^2} + \lambda_h\xi_h\frac{\partial^2 u_2}{\partial x_1 \partial x_2} + \lambda_h^2\frac{\partial^2 u_1}{\partial x_2^2} = 0. \tag{3.16}$$

Similarly, combining (3.13) and (3.14) it follows

$$\xi_h\lambda_h\frac{\partial^2 u_1}{\partial x_2 \partial x_1} + (\xi_h + \lambda_h)\lambda_h\frac{\partial^2 u_2}{\partial x_2^2} + \frac{\partial^2 u_2}{\partial x_1^2} = 0. \tag{3.17}$$

3.1 Fourier transform analysis

$$\mathcal{F}[f(t)](k) = \int_{-\infty}^{\infty} f(t)e^{-ikt}\,dt.$$

3.1.1 Fourier transform with respect to x_1:

$$\mathcal{F}_{x_1}[g(x_1, x_2)](k) = \hat{g}(x_2; k) = \int_{-\infty}^{\infty} g(x_1, x_2)e^{-ikx_1}\,dx_1,$$

We use the following notation

$$[g(x_1, x_2)]\widehat{\ } = \widehat{g}(x_2; k),$$

with the properties given by

$$
\begin{aligned}
&\left[\frac{\partial g}{\partial x_1}(x_1,x_2)\right]^{\widehat{}} = ik\widehat{g}(x_2;k),\\
&\left[\frac{\partial^2 g}{\partial x_1^2}(x_1,x_2)\right]^{\widehat{}} = \left[\frac{\partial}{\partial x_1}\left(\frac{\partial g}{\partial x_1}(x_1,x_2)\right)\right]^{\widehat{}} = ik\left[\frac{\partial g}{\partial x_1}(x_1,x_2)\right]^{\widehat{}} = -k^2\widehat{g}(x_2;k).
\end{aligned}
$$

Inductively, we can deduce that,

$$
\left[\frac{\partial^m g}{\partial x_1^m}(x_1,x_2)\right]^{\widehat{}} = (ik)^m\widehat{g}(x_2;k),\ \forall m\in\mathbb{Z}^+.
$$

3.1.2 Solving the ODE system:

By application of Fourier transform and by virtue of properties previously showed to equations (3.16)-(3.17), we obtain the following ODE system

$$
\begin{cases}
-k^2(1+\lambda_h\xi_h)\widehat{u_1}(x_2;k) + \lambda_h^2\dfrac{\partial^2\widehat{u_1}(x_2;k)}{\partial x_2^2} + ik\lambda_h\xi_h\dfrac{\partial\widehat{u_2}(k;x_2)}{\partial x_2} = 0,\\
-k^2\widehat{u_2}(x_2;k) + (\xi_h+\lambda_h)\lambda_h\dfrac{\partial^2\widehat{u_2}(x_2;k)}{\partial x_2^2} + ik\xi_h\lambda_h\dfrac{\partial\widehat{u_1}(x_2;k)}{\partial x_2} = 0.
\end{cases}
\tag{3.18}
$$

Substituting by $u=\widehat{u_1}(x_2;k), v=\widehat{u_2}(x_2;k), c_1=-k^2(1+\lambda_h\xi_h), c_2=\lambda_h^2, c_3=ik\xi_h\lambda_h, c_4=-k^2$ and $c_5=(\xi_h+\lambda_h)\lambda_h$ in (3.18), we have

$$
\begin{cases}
c_1u + c_2u'' + c_3v' = 0,\\
c_4v + c_5v'' + c_3u' = 0.
\end{cases}
\tag{3.19}
$$

By using the differential operator D and D^2 instead of $'$ and $''$ respectively, we have that (3.19) becomes into

$$
\begin{cases}
(D^2c_2+c_1)u + c_3Dv = 0,\\
(D^2c_5+c_4)v + c_3Du = 0.
\end{cases}
$$

$$
\iff \begin{cases}
(D^2c_2+c_1)u + (c_3D)v = 0,\\
(c_3D)u + (D^2c_5+c_4)v = 0.
\end{cases}
\tag{3.20}
$$

In order to eliminate variable u in (3.20) we can multiply by (c_3D) the first equation and the second one by $-(c_2D^2+c_1)$ and obtain,

$$
\begin{cases}
(c_3D)(D^2c_2+c_1)u + (c_3D)^2v = 0,\\
-(c_3D)(D^2c_2+c_1)u - (D^2c_2+c_1)(D^2c_5+c_4)v = 0,
\end{cases}
$$

therefore

$$
[(c_3D)^2 - (D^2c_2+c_1)(D^2c_5+c_4)]v = 0,\ \text{or}\ (c_3D)^2 - (D^2c_2+c_1)(D^2c_5+c_4) = 0.
$$

By algebraic manipulation we have,

$$\begin{aligned} c_3^2 - [c_2c_5D^4 + c_2c_4D^2 + c_1c_5D^2 + c_1c_4] &= -c_2c_5D^4 + (c_3^2 - c_2c_4 - c_1c_5)D^2 - c_1c_4 \\ &= c_2c_5D^4 + (c_1c_5 + c_2c_4 - c_3^2)D^2 + c_1c_4. \end{aligned}$$

Finally, we obtain

$$c_2c_5D^4 + (c_1c_5 + c_2c_4 - c_3^2)D^2 + c_1c_4 = 0. \tag{3.21}$$

Similarly, in order to eliminate variable v in (3.20),we can multiply by $(c_5D^2 + c_4)$ the first equation and the second one by $-(c_3D)$ and obtain,

$$\begin{cases} (D^2c_2 + c_1)(c_5D^2 + c_4)u + (c_3D)(c_5D^2 + c_4)v = 0, \\ -(c_3D)^2u - (c_3D)(D^2c_5 + c_4)v = 0, \end{cases}$$

therefore

$$[(D^2c_2 + c_1)(c_5D^2 + c_4) - (c_3D)^2]v = 0, \text{ or } (D^2c_2 + c_1)(c_5D^2 + c_4) - (c_3D)^2 = 0.$$

By algebraic manipulation we have,

$$\begin{aligned} (D^2c_2 + c_1)(c_5D^2 + c_4) - (c_3D)^2 &= c_2c_5D^4 + c_2c_4D^2 + c_1c_5D^2 + c_1c_2 - c_3D^2 \\ &= 0. \end{aligned}$$

Finally, we obtain

$$c_2c_5D^4 + (c_1c_5 + c_2c_4 - c_3^2)D^2 + c_1c_4 = 0. \tag{3.22}$$

Now, since (3.21)-(3.22) are the same, we can compute the coefficients as follows

$$\begin{aligned} c_2c_5 &= \lambda_h^3(\xi_h + \lambda_h), \qquad (3.23) \\ c_1c_4 &= k^4(1 + \lambda_h\xi_h), \qquad (3.24) \\ c_1c_5 &= -k^2(1 + \lambda_h\xi_h)\lambda_h(\xi_h + \lambda_h) = -k^2\lambda_h(\xi_h + \lambda_h + \lambda_h\xi_h^2 + \lambda_h^2\xi_h), \\ c_2c_4 &= -k^2\lambda_h, \\ -c_3^2 &= k^2\lambda_h^2\xi_h^2, \end{aligned}$$

then,

$$\begin{aligned} c_1c_5 + c_2c_5 - c_3^2 &= -k^2\lambda_h\xi_h - k^2\lambda_h^2 - k^2\lambda^2\xi_h^2 - k^2\lambda_h^3\xi_h - k^2\lambda_h^2 + k^2\lambda_h^2\xi_h^2 \\ &= -(\xi_h + 2\lambda_h + \lambda_h^2\xi_h)\lambda_hk^2. \end{aligned} \tag{3.25}$$

Substituting (3.23)-(3.25) into (3.22), we have

$$\lambda_h^3(\xi_h + \lambda_h)D^4 - (\xi_h + 2\lambda_h + \lambda_h^2\xi_h)\lambda_hk^2D^2 + (1 + \lambda_h\xi_h)k^4 = 0, \tag{3.26}$$

Setting $\lambda_h = a$ and $\xi_h = b$ and $D = q$ we obtain the following equation

$$a^3(a + b)q^4 - (b + 2a + a^2b)ak^2q^2 + (1 + ab)k^4 = 0,$$

$$q^2 = \frac{(b + 2a + a^2b)ak^2 \pm \sqrt{\Delta}}{2a^3(a + b)},$$

where,

$$
\begin{aligned}
\Delta &= [(b+2a+a^2b)ak^2]^2 - 4a^3(a+b)(1+ab)k^4 \\
&= (b+2a+a^2b)^2a^2k^4 - 4a^3k^4(a+b)(1+ab) \\
&= [b^4 + 2b(2a+a^2b) + (2a+a^2b)^2]a^2k^4 - 4a^3k^4(a+a^2b+b+ab^2) \\
&= [b^2 + 4ab + 2a^2b^2 + 4a^2 + 4a^3b + a^4b^2]a^2k^4 - 4a^4k^4 - 4a^5bk^4 - 4a^3bk^4 - 4a^4b^2k^4 \\
&= a^2b^2k^4 + 4a^3bk^4 + 2a^4b^2k^4 + 4a^4k^4 + 4a^5bk^4 + a^6b^2k^4 - 4a^4k^4 - 4a^5bk^4 - 4a^3bk^4 - 4a^4b^2k^4 \\
&= a^6b^2k^4 + a^2b^2k^4 - 2a^4b^2k^4 = a^2b^2k^4(a^4 + 1 - 2a^2) \\
&= a^2b^2k^4(a^2-1)^2
\end{aligned}
$$

therefore,

$$
q^2 = \frac{(b+2a+a^2b)ak^2 \pm abk^2(a^2-1)}{2a^3(a+b)} = \frac{[b+2a+a^2b \pm (a^2b-b)]ak^2}{2a^3(a+b)},
$$

now,

$$
\begin{cases}
q_{1,2}^2 = \dfrac{[b+2a+a^2b+(a^2b-b)]ak^2}{2a^3(a+b)} = \dfrac{[\cancel{b}+2a+a^2b+a^2b\cancel{-b}]ak^2}{2a^3(a+b)} \\
q_{3,4}^2 = \dfrac{[b+2a+a^2b-(a^2b-b)]ak^2}{2a^3(a+b)} = \dfrac{[b+2a+\cancel{a^2b-a^2b}+b]ak^2}{2a^3(a+b)}
\end{cases}
$$

$$
\begin{cases}
q_{3,4}^2 = \dfrac{[2a+2a^2b]ak^2}{2a^3(a+b)} = \dfrac{2a(1+ab)ak^2}{2a^3(a+b)} = k^2\dfrac{(1+ab)}{a(a+b)}, \\
q_{1,2}^2 = \dfrac{[2b+2a]ak^2}{2a^3(a+b)} = \dfrac{2(a+b)ak^2}{2a^3(a+b)} = \dfrac{k^2}{a^2},
\end{cases}
$$

$$
\begin{cases}
q_{3,4} = \pm k\sqrt{\dfrac{(1+ab)}{a(a+b)}}, \\
q_{1,2} = \pm \dfrac{k}{a}.
\end{cases}
$$

Finally,

$$
\begin{cases}
q_{3,4} = \pm k\sqrt{\dfrac{(1+\lambda_h\xi_h)}{\lambda_h(\lambda_h+\xi_h)}} = \pm k\beta, \quad \text{where } \beta = \sqrt{\dfrac{(1+\lambda_h\xi_h)}{\lambda_h(\lambda_h+\xi_h)}}, \\
q_{1,2} = \pm \dfrac{k}{\lambda_h}.
\end{cases}
\tag{3.27}
$$

Degenerate states

Degenerate states results when two(o more) different eigenstates to an eigenvalue equation correspond to the same eigen value. In our problem we have two degenerations:

(a) $\xi_h = 0$,

(b) $\beta = \sqrt{\dfrac{(1+\lambda_h\xi_h)}{\lambda_h(\lambda_h+\xi_h)}} = 0$ or equivalently $\lambda_h\xi_h = -1$.

Case $\xi_h = 0$: The equation

$$\lambda_h^3(\xi_h + \lambda_h)q^4 - (\xi_h + 2\lambda_h + \lambda_h^2\xi_h)\lambda_h k^2 q^2 + (1 + \lambda_h\xi_h)k^4 = 0,$$

reduces to

$$\lambda_h^3\lambda_h q^4 - 2\lambda_h^2 k^2 q^2 + k^4 = 0 \Longrightarrow (\lambda_h^2 q^2 - k^2)^2 = 0,$$

therefore,

$$(\lambda_h q - k)^2(\lambda_h q + k)^2 = 0,$$

then

$$q_{1,2} = \pm\frac{k}{\lambda_h}.$$

Case $\lambda_h\xi_h = -1$: The equation

$$\lambda_h^3(\xi_h + \lambda_h)q^4 - (\xi_h + 2\lambda_h + \lambda_h^2\xi_h)\lambda_h k^2 q^2 + (1 + \lambda_h\xi_h)k^4 = 0,$$

reduces to

$$(\lambda_h^4 - \lambda_h^2)q^4 - (-1 + 2\lambda_h^2 - \lambda_h^2)k^2 q^2 + (1-1)k^4 = 0 \Longrightarrow (\lambda_h^4 - \lambda_h^2)q^4 - (-1 + 2\lambda_h^2 - \lambda_h^2)k^2 q^2 = 0$$

therefore,

$$-\lambda_h^2(\lambda_h^2 - 1)q^4 - (-\lambda_h^2 + 1)k^2 q^2 = 0 \Longrightarrow -\lambda_h^2 q^4 + k^2 q^2 = 0,$$

then

$$q^2(-\lambda_h^2 q^2 + k^2) = 0 \Longrightarrow q^2(k - \lambda_h q)(k + \lambda_h q) = 0,$$

finally,

$$\begin{cases} q_{1,2} = 0, \\ q_{3,4} = \pm\dfrac{k}{\lambda_h}. \end{cases}$$

3.2 Solution of ODE (3.18)

We know by virtue of standard ODE theory that the solution of the system (3.18) has the form

$$\widehat{u_1}(x_2; k) = \sum_{n=1}^{4} \overline{u_1^{(n)}} e^{q_n x_2}, \tag{3.28}$$

$$\widehat{u_2}(x_2; k) = \sum_{n=1}^{4} \overline{u_2^{(n)}} e^{q_n x_2}, \tag{3.29}$$

where q_n are as in (3.27) and $\overline{u_1^n}, \overline{u_2^n}$ are constants. In order to compute $\overline{u_1^n}, \overline{u_2^n}$ we can substituting (3.28)-(3.29) into (3.18) and we obtain

$$\begin{cases} [-k^2(1 + \lambda_h\xi_h) + \lambda_h^2 q_n^2]\overline{u_1^{(n)}} + ik\lambda_h\xi_h q_n \overline{u_2^{(n)}} = 0, \\ ik\lambda_h\xi_h q_n \overline{u_1^{(n)}} + [-k^2 + (\xi_h + \lambda_h)\lambda_h q_n^2]\overline{u_2^{(n)}} = 0. \end{cases} \tag{3.30}$$

By substituting $n = 1$ and $q_1 = \frac{k}{\lambda_h}$ into (3.30), we have

$$\begin{cases} [-k^2(1+\lambda_h\xi_h)+k^2]\overline{u_1^{(1)}} + ik\xi_h\overline{u_2^{(1)}} = 0, \\ ik^2\xi_h\overline{u_1^{(1)}} + [-k^2 + \frac{k^2}{\lambda_h}(\xi_h+\lambda_h)]\overline{u_2^{(1)}} = 0, \end{cases}$$

$$\Longrightarrow \begin{cases} -k^2\lambda_h\xi_h\overline{u_1^{(1)}} + ik\xi_h\overline{u_2^{(1)}} = 0, \\ ik^2\xi_h\overline{u_1^{(1)}} + k^2\frac{\xi_h}{\lambda_h}\overline{u_2^{(1)}} = 0. \end{cases} \tag{3.31}$$

We can see that first equation in (3.31) is $\lambda_h i$ times the second one equation, therefore we have infinity solutions and they are of the form

$$\overline{u_1^{(1)}} = A_1,\ \overline{u_2^{(1)}} = -i\lambda_h A_1,\ \text{where} A_1 \text{ is constant.} \tag{3.32}$$

By substituting $n = 2$ and $q_2 = -\frac{k}{\lambda_h}$ into (3.30), we have

$$\begin{cases} [-k^2(1+\lambda_h\xi_h)+k^2]\overline{u_1^{(2)}} - ik\xi_h\overline{u_2^{(2)}} = 0, \\ -ik^2\xi_h\overline{u_1^{(2)}} + [-k^2 + \frac{k^2}{\lambda_h}(\xi_h+\lambda_h)]\overline{u_2^{(2)}} = 0, \end{cases}$$

$$\Longrightarrow \begin{cases} -k^2\lambda_h\xi_h\overline{u_1^{n}} - ik\xi_h\overline{u_2^{n}} = 0, \\ -ik^2\xi_h\overline{u_1^{n}} + k^2\frac{\xi_h}{\lambda_h}\overline{u_2^{n}} = 0. \end{cases} \tag{3.33}$$

We can see that first equation in (3.33) is $-\lambda_h i$ times the second one equation, therefore we have infinity solutions and they are of the form

$$\overline{u_1^{(2)}} = A_2,\ \overline{u_2^{(2)}} = i\lambda_h A_2,\ \text{where} A_2 \text{ is constant.} \tag{3.34}$$

By substituting $n = 3$ and $q_3 = k\beta$ into (3.30), we have

$$\begin{cases} [-k^2(1+\lambda_h\xi_h)+k^2\lambda_h^2\beta^2]\overline{u_1^{(3)}} + ik^2\beta\lambda_h\xi_h\overline{u_2^{(3)}} = 0, \\ ik^2\beta\lambda_h\xi_h\overline{u_1^{(3)}} + [-k^2 + \lambda_h k^2\beta^2(\xi_h+\lambda_h)]\overline{u_2^{(3)}} = 0, \end{cases}$$

let $a = -k^2 + k^2\lambda_h^2\beta^2, b = k^2\lambda_h\xi_h, r = ik^2\beta\lambda_h\xi_h = ib\beta, x = \overline{u_1^{(3)}}$ and $y = \overline{u_2^{(3)}}$, then we obtain

$$\begin{cases} (a-b)x + ib\beta y = 0, \\ ib\beta x + (a+b\beta^2)y = 0, \end{cases}$$

multiplying by $-i\beta$ the first equation we have

$$\begin{cases} (a\beta i - ib\beta)x - b\beta^2 y = 0, \\ ib\beta x + (a+b\beta^2)y = 0. \end{cases}$$

By addition of the second one equation with the first, we have

$$\begin{cases} a\beta ix - ay = 0, \\ ib\beta x + (a+b\beta^2)y = 0. \end{cases} \tag{3.35}$$

We can see that equations in (3.35) are the same, then by virtue of first equation we have

$$\beta ix + y = 0,$$

therefore, we have infinity solutions and the are of the form

$$\overline{u_1^{(3)}} = A_3,\ \overline{u_2^{(3)}} = -i\beta A_3,\ \text{where} A_3 \text{ is constant.} \tag{3.36}$$

By substituting $n = 4$ and $q_4 = -k\beta$ into (3.30), we have

$$\begin{cases} [-k^2(1+\lambda_h\xi_h) + k^2\lambda_h^2\beta^2]\overline{u_1^{(4)}} - ik^2\beta\lambda_h\xi_h\overline{u_2^{(4)}} = 0, \\ -ik^2\beta\lambda_h\xi_h\overline{u_1^{(4)}} + [-k^2 + \lambda_h k^2\beta^2(\xi_h+\lambda_h)]\overline{u_2^{(4)}} = 0, \end{cases}$$

let $a = -k^2 + k^2\lambda_h^2\beta^2, b = k^2\lambda_h\xi_h, r = ik^2\beta\lambda_h\xi_h = ib\beta, x = \overline{u_1^{(3)}}$ and $y = \overline{u_2^{(3)}}$, then we obtain

$$\begin{cases} (a-b)x - ib\beta y = 0, \\ -ib\beta x + (a+b\beta^2)y = 0, \end{cases}$$

multiplying by $-i\beta$ the first equation we have

$$\begin{cases} (-a\beta i + ib\beta)x - b\beta^2 y = 0, \\ -ib\beta x + (a+b\beta^2)y = 0. \end{cases}$$

By addition of the second one equation with the first, we have

$$\begin{cases} -a\beta ix + ay = 0, \\ -ib\beta x + (a+b\beta^2)y = 0. \end{cases} \tag{3.37}$$

We can see that equations in (3.37) are the same, then by virtue of first equation we have

$$-\beta ix + y = 0,$$

therefore, we have infinity solutions and the are of the form

$$\overline{u_1^{(4)}} = A_4,\ \overline{u_2^{(4)}} = i\beta A_4,\ \text{where} A_4 \text{ is constant.} \tag{3.38}$$

Finally, by virtue of (3.32), (3.34), (3.36) and (3.38) the solutions (3.28)-(3.29) can be rewritten as follows

$$\widehat{u_1}(x_2;k) = \sum_{n=1}^{4} A_n \overline{u}_1^{(n)} e^{q_n x_2}, \tag{3.39}$$

$$\widehat{u_2}(x_2;k) = \sum_{n=1}^{4} A_n \overline{u}_2^{(n)} e^{q_n x_2}, \tag{3.40}$$

where,

$$\begin{bmatrix} \overline{u}_1^{(1)} & \overline{u}_1^{(2)} & \overline{u}_1^{(3)} & \overline{u}_1^{(4)} \\ \overline{u}_2^{(1)} & \overline{u}_2^{(2)} & \overline{u}_2^{(3)} & \overline{u}_2^{(4)} \end{bmatrix} = \begin{bmatrix} 1 & 1 & 1 & 1 \\ -i\lambda_h & i\lambda_h & -i\beta & i\beta \end{bmatrix}. \tag{3.41}$$

3.2.1 Boundary condition

The boundary condition

$$P : N = T \,\text{on}\, \Gamma_0. \tag{3.42}$$

In the other hand, the upper surface of the hydrogel is subjected to a normal traction due to the pressure of external solvent p, i.e.

$$\sigma n \, da = -pn \, da. \tag{3.43}$$

We will try to stablish a relationship between (3.42) and (3.43), as follows:

$$\vec{F} = -p \cdot (\operatorname{Cof} F) N \, dA,$$

$$P : N \, dA = -p \cdot (\operatorname{Cof} F) N \, dA,$$

therefore,

$$P : N = \underbrace{-p \cdot (\operatorname{Cof} F) N}_{T},$$

in the other hand

$$n \, da = (\operatorname{Cof} F) N \, dA,$$

for $N = \begin{pmatrix} 0 \\ 1 \\ 0 \end{pmatrix}$, then

$$T = -p(\operatorname{Cof} F) N = -p(\operatorname{Cof} F) \begin{pmatrix} 0 \\ 1 \\ 0 \end{pmatrix}$$

this imply that

$$T = -p \cdot \begin{pmatrix} (\operatorname{Cof} F)_{12} \\ (\operatorname{Cof} F)_{22} \\ (\operatorname{Cof} F)_{32} \end{pmatrix}, \tag{3.44}$$

and

$$P : N = \begin{pmatrix} P_{12} \\ P_{22} \\ P_{32} \end{pmatrix}. \tag{3.45}$$

Finally, from (3.42), (3.44) and (3.45) we deduce that

$$\begin{aligned} P_{12} &= -p(\operatorname{Cof} F)_{12} = p\partial_1 u_2 \,\text{on}\, x_2 = h, \\ P_{22} &= -p(\operatorname{Cof} F)_{22} = -p(1 + \partial_1 u_1) \,\text{on}\, x_2 = h. \end{aligned}$$

In our case $p = 0$, therefore the boundary condition are:

$$P_{12} = 0 \,\text{on}\, x_2 = h, \tag{3.46}$$

$$P_{22} = 0 \,\text{on}\, x_2 = h. \tag{3.47}$$

The boundary condition $u_1(x_1, 0) = u_2(x_1, 0) = 0$, implies that

$$\widehat{u}_1(0, k) = \widehat{u}_2(0, k) = 0. \tag{3.48}$$

By application of (3.48) in (3.39)-(3.40) we obtain

$$\sum_{n=1}^{4} A_n \overline{u}_1^{(n)} = 0, \tag{3.49}$$

$$\sum_{n=1}^{4} A_n \overline{u}_2^{(n)} = 0. \tag{3.50}$$

In the other hand by (3.47) we have,

$$P_{22} = 0 \quad \text{at } x_2 = h,$$

then

$$\left[(\xi_h - \lambda_h)\frac{\partial u_1}{\partial x_1} + (\xi_h + \lambda_h)\frac{\partial u_2}{\partial x_2}\right] = 0 \quad \text{at } x_2 = h. \tag{3.51}$$

By application of Fourier transformation, we have

$$ik(\xi_h - \lambda_h)\widehat{u}_1 + (\xi_h + \lambda_h)\frac{\partial \widehat{u}_2}{\partial x_2} = 0 \quad \text{at } x_2 = h. \tag{3.52}$$

Now, by application of (3.39)-(3.40) in (3.52) we obtain

$$\sum_{n=1}^{4} \left[ik(\xi_h - \lambda_h)\overline{u}_1^{(n)} + (\xi_h + \lambda_h)q_n \overline{u}_2^{(n)}\right] A_n e^{q_n h} = 0. \tag{3.53}$$

Having in mind the first boundary condition (3.46)

$$P_{12} = 0 \quad \text{at } x_2 = h,$$

then,

$$\lambda_h \left[\frac{\partial u_1}{\partial x_2} + \frac{\partial u_2}{\partial x_1}\right] = 0 \quad \text{at } x_2 = h.$$

By application of Fourier transformation, we have

$$\lambda_h \left[\frac{\partial \widehat{u}_1}{\partial x_2} + ik\widehat{u}_2\right] = 0 \quad \text{at } x_2 = h. \tag{3.54}$$

Now, by application of (3.39)-(3.40) in (3.53) we obtain

$$\sum_{n=1}^{4} \left[\lambda_h q_n \overline{u}_1^{(n)} + ik\lambda_h \overline{u}_2^{(n)}\right] A_n e^{q_n h} = 0. \tag{3.55}$$

3.2.2 Calculation of normal n

Since

$$n\,da = (\mathrm{Cof})N\,dA,$$

then

$$n = \frac{(\mathrm{Cof}\,F)N}{|(\mathrm{Cof}\,F)N|} = \frac{(\mathrm{Cof}\,F)\begin{pmatrix}0\\1\\0\end{pmatrix}}{\left|(\mathrm{Cof}\,F)\begin{pmatrix}0\\1\\0\end{pmatrix}\right|} = \frac{\begin{pmatrix}(\mathrm{Cof}\,F)_{12}\\(\mathrm{Cof}\,F)_{22}\\(\mathrm{Cof}\,F)_{32}\end{pmatrix}}{\left|\begin{pmatrix}(\mathrm{Cof}\,F)_{12}\\(\mathrm{Cof}\,F)_{22}\\(\mathrm{Cof}\,F)_{32}\end{pmatrix}\right|} = \begin{pmatrix}-\frac{\partial_1 u_2}{\sqrt{(1+\partial_1 u_1)^2+\partial_1 u_2^2}}\\(1+\frac{\partial_1 u_2^2}{(1+\partial_1 u_1)^2})^{-1/2}\\0\end{pmatrix},$$

therefore

$$n \approx \begin{pmatrix}-\frac{\partial_1 u_2}{1}\\1-\frac{1}{2}\frac{\partial_1 u_2^2}{(1+\partial_1 u_1)^2}\\0\end{pmatrix} \approx \begin{pmatrix}-\partial_1 u_2\\1-\frac{1}{2}\partial_1 u_2^2\\0\end{pmatrix}. \tag{3.56}$$

Finally,

$$n_1 \approx -\partial_1 u_2,$$
$$n_2 \approx 1-\frac{1}{2}\partial_1 u_2^2.$$

3.2.3 Calculation of A_n

From equations (3.49), (3.50), (3.53) and (3.55) we have the following 4×4 system where A_n are the four unknowns

$$\sum_{n=1}^{4} A_n \overline{u}_1^{(n)} = 0, \tag{3.57}$$

$$\sum_{n=1}^{4} A_n \overline{u}_2^{(n)} = 0, \tag{3.58}$$

$$\sum_{n=1}^{4}\left[ik(\xi_h-\lambda_h)\overline{u}_1^{(n)} + (\xi_h+\lambda_h)q_n\overline{u}_2^{(n)}\right]A_n e^{q_n h} = 0, \tag{3.59}$$

$$\sum_{n=1}^{4}\left[\lambda_h q_n\overline{u}_1^{(n)} + ik\lambda_h\overline{u}_2^{(n)}\right]A_n e^{q_n h} = 0. \tag{3.60}$$

where, $\overline{u}_i^{(n)}$ with $i\in\{1,2\}$ are as in (3.73).

We define

$$\sum_{n=1}^{4} D_{mn}(k)A_n = 0. \tag{3.61}$$

From (3.57), we have

$$A_1 + A_2 + A_3 + A_4 = 0, \tag{3.62}$$

therefore

$$D_{11}(k) = D_{12}(k) = D_{13}(k) = D_{14}(k) = 1. \tag{3.63}$$

Multiplying by i^{-1} equation (3.57) we obtain

$$-\lambda_h A_1 + \lambda_h A_2 - \beta A_3 + \beta A_4 = 0, \tag{3.64}$$

therefore

$$D_{21}(k) = -\lambda_h, D_{22}(k) = \lambda_h, D_{23}(k) = -\beta \text{ and } D_{24}(k) = \beta. \tag{3.65}$$

From (3.59) and $\lambda_h = \dfrac{h}{h_0}$, we have

$$\begin{aligned} D_{31}(k) &= [ik(\xi_h - \lambda_h) + (\xi_h + \lambda_h)\frac{k}{\lambda_h}(-i\lambda_h)]e^{\frac{k}{\lambda_h}h} \\ &= [ik(\xi_h - \lambda_h) + (\xi_h + \lambda_h)(-ki)]e^{\frac{k}{\lambda_h}h} \\ &= [ik(\xi_h - \lambda_h) - ki\xi_h - k\lambda_h i]e^{kh_0} \\ &= -2k\lambda_h ie^{kh_0}. \end{aligned} \tag{3.66}$$

$$\begin{aligned} D_{32}(k) &= \left[ik(\xi_h - \lambda_h) + (\xi_h + \lambda_h)(i\lambda_h)\left(\frac{-k}{\lambda_h}\right)\right] e^{-\frac{k}{\lambda_h}h} \\ &= [ik\xi_h - ik\lambda_h - ik\xi_h - ik\lambda_h]e^{-kh_0} \\ &= -2k\lambda_h ie^{-kh_0}. \end{aligned} \tag{3.67}$$

$$\begin{aligned} D_{33}(k) &= [ik(\xi_h - \lambda_h) + (\xi_h + \lambda_h)k\beta(-i\beta)]e^{k\beta h} \\ &= [ik(\xi_h - \lambda_h) + [(\xi_h + \lambda_h)\beta^2](-ki)]e^{k\beta h} \\ &= \left[(\lambda_h - \xi_h) + \left(\frac{1 + \lambda_h\xi_h}{\lambda_h}\right)\right](-ki)e^{k\beta h} \\ &= -ki\left(\lambda_h + \frac{1}{\lambda_h}\right)e^{k\beta h}. \end{aligned} \tag{3.68}$$

$$\begin{aligned} D_{43}(k) &= [ik(\xi_h - \lambda_h) + (\xi_h + \lambda_h)(-k\beta)(i\beta)]e^{-k\beta h} \\ &= [(\lambda_h - \xi_h) + (\xi_h + \lambda_h)\beta^2](-ki)e^{-k\beta h} \\ &= -ki\left(\lambda_h + \frac{1}{\lambda_h}\right)e^{-k\beta h}. \end{aligned} \tag{3.69}$$

From, (3.66)-(3.69) and since $k \neq 0$ then

$$-2k\lambda_h ie^{kh_0}A_1 - 2k\lambda_h ie^{-kh_0}A_2 - ki\left(\lambda_h + \frac{1}{\lambda_h}\right)e^{k\beta h}A_3 - ki\left(\lambda_h + \frac{1}{\lambda_h}\right)e^{-k\beta h}A_4 = 0.$$

Finally, multiplying by $-\frac{1}{ki}$ we have,

$$2\lambda_h e^{kh_0} A_1 + 2\lambda_h e^{-kh_0} A_2 + \left(\lambda_h + \frac{1}{\lambda_h}\right) e^{k\beta h} A_3 + \left(\lambda_h + \frac{1}{\lambda_h}\right) e^{-k\beta h} A_4 = 0. \tag{3.70}$$

Similarly, we can calculate $D_{4j}(k)$ for $j \in \{1, 2, 3, 4\}$ by virtue of equation (3.60) and we obtain

$$D_{41}(k) = \left(\lambda_h + \frac{1}{\lambda_h}\right) e^{kh_0}, D_{42}(k) = -\left(\lambda_h + \frac{1}{\lambda_h}\right) e^{-kh_0}, D_{43}(k) = 2\beta e^{\beta kh}, \tag{3.71}$$

$$D_{44}(k) = -2\beta e^{-\beta kh}. \tag{3.72}$$

From the calculations of $D_{mn}(k)$ we have the matrix

$$[D_{mn}(k)] = \begin{bmatrix} 1 & 1 & 1 & 1 \\ -\lambda_h & \lambda_h & -\beta & \beta \\ 2\lambda_h e^{kh_0} & 2\lambda_h e^{-kh_0} & \left(\lambda_h + \frac{1}{\lambda_h}\right) e^{k\beta h} & \left(\lambda_h + \frac{1}{\lambda_h}\right) e^{-k\beta h} \\ \left(\lambda_h + \frac{1}{\lambda_h}\right) e^{kh_0} & -\left(\lambda_h + \frac{1}{\lambda_h}\right) e^{-kh_0} & 2\beta e^{\beta kh} & -2\beta e^{-\beta kh} \end{bmatrix}. \tag{3.73}$$

Let $a = \lambda_h$, $b = \beta$, $c = h_0$ and $d = h$, we have

$$[D_{mn}(k)] = \begin{bmatrix} 1 & 1 & 1 & 1 \\ -a & a & -b & b \\ 2ae^{ck} & 2ae^{-ck} & \left(a + \frac{1}{a}\right) e^{bdk} & \left(a + \frac{1}{a}\right) e^{-bdk} \\ \left(a + \frac{1}{a}\right) e^{ck} & -\left(a + \frac{1}{a}\right) e^{-ck} & 2be^{bdk} & -2be^{-bdk} \end{bmatrix}. \tag{3.74}$$

We have that the unstable condition is

$$\det[D_{mn}] = f(kh_0, \lambda_h; N\nu, \chi, \phi_0) = 0, \tag{3.75}$$

therefore,

$$e^{bdk} \left[\frac{e^{-ck}(a+b)\left(a^4 + 4a^3b + 2a^2 + 1\right) e^{ck} - e^{ck}(a^5 - 5a^4b + 2a^3(2b^2+1) - 2a^2b + a - b)}{a^2} \right]$$
$$+ e^{-bdk} \left[\frac{e^{ck}(a+b)\left(a^4 + 4a^3b + 2a^2 + 1\right) + e^{-ck}(a^4 - 4a^3b + 2a^2 + 1)(b - a)}{a^2} \right] - 16b(a^2+1) = 0,$$

i.e.

$$e^{\beta hk} \left[\frac{e^{-h_0k}(\lambda_h + \beta)\left(\lambda_h^4 + 4\lambda_h^3\beta + 2\lambda_h^2 + 1\right) e^{h_0k} - e^{h_0k}(\lambda_h^5 - 5\lambda_h^4\beta + 2\lambda_h^3(2\beta^2+1) - 2\lambda_h^2\beta + \lambda_h - \beta)}{\lambda_h^2} \right]$$
$$+ e^{-\beta hk} \left[\frac{e^{h_0k}(\lambda_h + \beta)\left(\lambda_h^4 + 4\lambda_h^3\beta + 2\lambda_h^2 + 1\right) + e^{-h_0k}(\lambda_h^4 - 4\lambda_h^3\beta + 2\lambda_h^2 + 1)(\beta - \lambda_h)}{\lambda_h^2} \right]$$
$$- 16\beta(\lambda_h^2 + 1) = 0.$$

By substituting $z = h_0 k$, we have

$$e^{\lambda_h \beta z}\left[\frac{e^{-z}(\lambda_h+\beta)\left(\lambda_h^4+4\lambda_h^3\beta+2\lambda_h^2+1\right)e^z - e^z(\lambda_h^5-5\lambda_h^4\beta+2\lambda_h^3(2\beta^2+1)-2\lambda_h^2\beta+\lambda_h-\beta)}{\lambda_h^2}\right]$$
$$+\,e^{-\beta\lambda_h z}\left[\frac{e^z(\lambda_h+\beta)\left(\lambda_h^4+4\lambda_h^3\beta+2\lambda_h^2+1\right)+e^{-z}(\lambda_h^4-4\lambda_h^3\beta+2\lambda_h^2+1)(\beta-\lambda_h)}{\lambda_h^2}\right]$$
$$-\,16\beta(\lambda_h^2+1)=0.$$

For each kh_0, Eq.(3.75) predicts a critical swelling ratio, $\lambda_c(kh_0; N\nu, \chi)$, which depends on the perturbation wave number kh_0 as well as the two material parameters ($N\nu$ and χ) of the hydrogel. The corresponding critical chemical potential (μ_c) can the be determinated from the homogeneous solution Eq.(2.6).

Fig.1 plots the predicted critical swelling ratio as a function of kh_0, and Fig.2 plots the critical chemical potential, for $\overline{p}_0 = 2.3\times10^{-5}$, $N\nu = 0.001$, and differents values of χ. Unlike the critical compression surface instability of a semi-infinite rubber, which is independent of the perturbation wavelength, the critical swelling ratio for swell-induced surface instability of a hydrogel layer depends on the normalized wave number, kh_0 due the presence of a rigid substrate. The substrate confinement tends to stabilize long-wavelength perturbations (with small kh_0), while the confinement effect diminishes for short-wavelength perturbations (with large kh_0). Consequently, the critical swelling ratio decreases as kh_0 increases and approaches a constan at the limit of short-wavelength perturbations ($kh_0 \to \infty$).

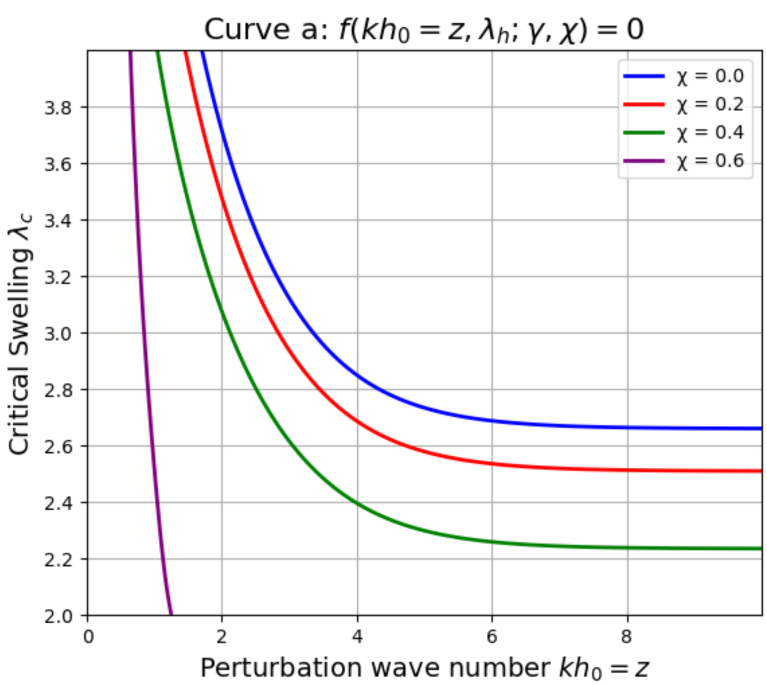

Figure 1: λ_c vs kh_0

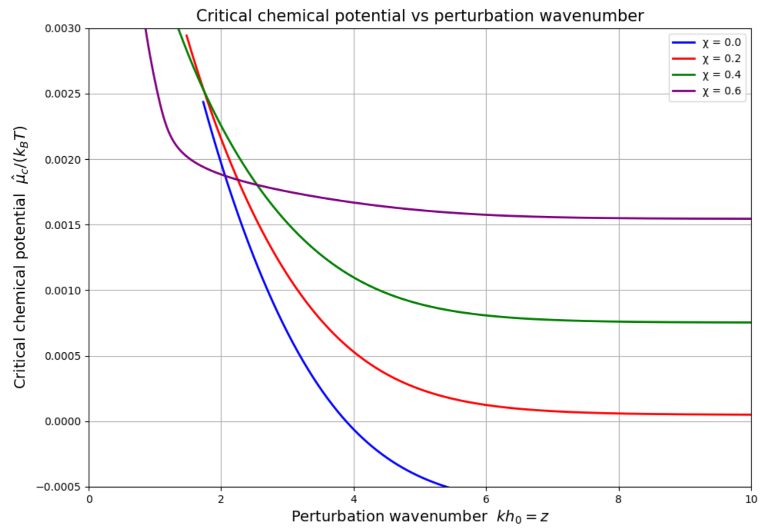

Figure 2: $\mu_c/(K_BT)$ vs kh_0

Therefore, the onset of swell-induced surface instability is controlled by the minimum critical swelling ratio at the short-wavelength limit. It is speculated that surface effects (e.g., surface energy, surface stress), while not considered in present study, could potentially stabilize short-wavelength perturbatins and, together with the substrate confinement effect, could lead to an intermediate wavelength for the onset of the surface instability. We note in Fig 4 that, for $\chi \leq 0.6$, there exists a critical wave number, for which the critical swelling ratio equals the maximum homogeneous swelling ratio and the corresponding critical chemical potential equals zero. For smaller perturbation wave numbers, the hydrogel layer remains stable at the equilibrium chemical potential ($\mu = \hat{\mu} = 0$). For $\chi > 0.6$, however, we find that the hydrogel layer remains stable for all posible perturbation wave numbers; thus no critical condition is predicted. As shows later (Fig 6), for each $N\nu$, there exists a critical value for χ, beyond which the hydrogel layer is stable as swells homogeneously at the equilibrium state.

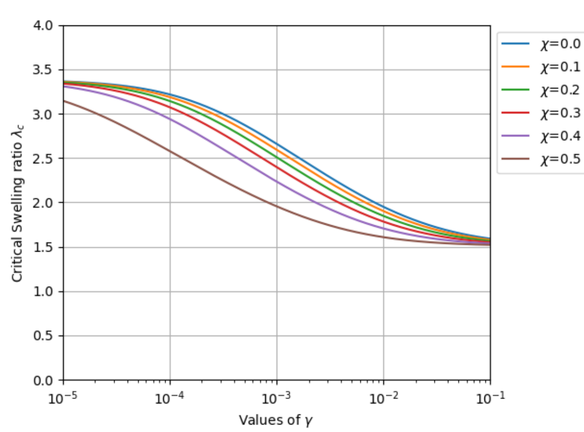

Figure 3: Critical swelling ratio

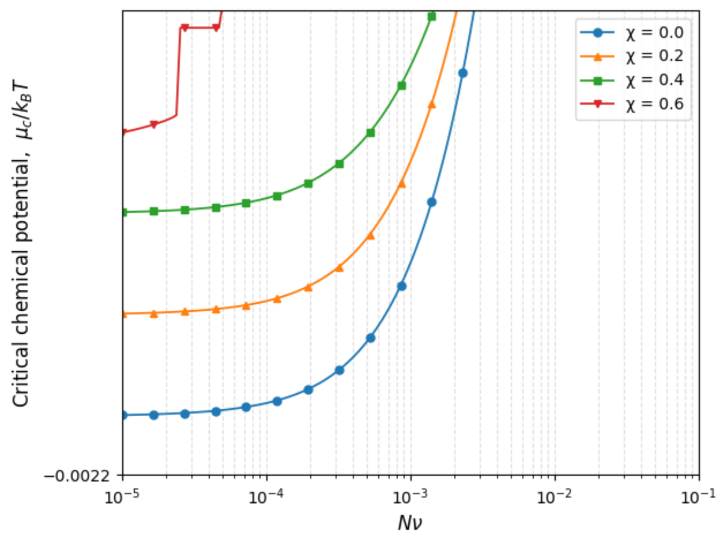

Figure 4: Critical chemical potential

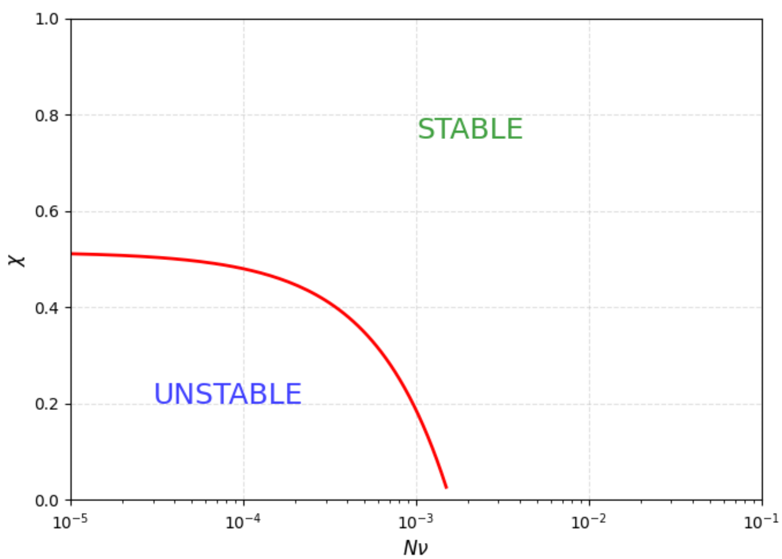

Figure 5: A stability diagram for substrate-confined hydrogel layer

Short-wavelength

If $ck \to \infty$ in (3.74) and since $e^{-bdk} = e^{-ck\left(\frac{b}{c}d\right)}$ we have

$$[D_{mn}(k)] = \begin{bmatrix} 1 & 1 & 1 & 1 \\ -a & a & -b & b \\ 2ae^{ck} & 0 & \left(a+\frac{1}{a}\right)e^{bdk} & 0 \\ \left(a+\frac{1}{a}\right)e^{ck} & 0 & 2be^{bdk} & 0 \end{bmatrix}. \tag{3.76}$$

Then, the unstable condition

$$\det[D_{mn}(k)] = 0,$$

is equivalent to

$$-e^{k(c+bd)}\left(a^5 - 5a^4 b + 4a^3 b^2 + 2a^3 - 2a^2 b + a - b\right)\frac{1}{a^2} = 0, \tag{3.77}$$

which is equivalent,

$$\left[\left(a + \frac{1}{a}\right)^2 - 4ab\right](b - a) = 0.$$

Next we focus our attention on the critical condition at the short-wavelength limit. By letting $kh_0 \to \infty$ in Eq.(3.72) and the setting the determinant of the matrix to be zero, we obtain that

$$f_\infty(\lambda_h; N\nu, \chi, \phi_0) = \left[\left(\lambda_h + \frac{1}{\lambda_h}\right)^2 - 4\lambda_h \beta\right](\beta - \lambda_h) = 0, \tag{3.78}$$

It can be shown that $\beta - \lambda_h \neq 0$ for all swelling hydrogels ($\lambda_h > 1$). Thus the critical condition becomes

$$\left(\lambda_h + \frac{1}{\lambda_h}\right)^2 - 4\lambda_h \beta = 0. \tag{3.79}$$

Combining this with the definition of β in (3.27) and ξ_h in (3.4) gives a nonlinear equation, which can be solved to predict the critical swelling ratio at the short-wavelength limit $\lambda_c^\infty(N\nu, \chi, \phi_0)$. The corresponding critical chemical potential, $\mu_c^\infty(N\nu, \chi, \overline{p}_0)$, is calculated from Eq.(2.6) by setting $\lambda_h = \lambda_c^\infty(N\nu, \chi, \phi_0)$.

Fig. 3 plots the predicted critical swelling ratio (λ_c^∞) as a function of $N\nu$ for different values of χ, and Fig. 4 plots the critical chemical potential (μ_c^∞), assuming a constant equilibrium vapor pressure ($\bar{p}_0 = 2.3 \times 10^{-5}$). The dashed lines in Fig. 3 show the maximum homogeneous swelling ratio predicted by Eq.(2.6). For each χ, the critical chemical potential increases monotonically with increasing Nv until it reaches the equilibrium chemical potential ($\mu_c^\infty = 0$), at which point the critical swelling ratio equals the maximum homogeneous swell ratio. Therefore, the stability of the homogeneously swollen hydrogel layer depends on both $N\nu$ and χ. The two dimensional material parameters characterize the elastic stiffness of the polymer network and the polymer–solvent interaction, respectively. While the polymer stiffness increases with Nv, the network tends to swell more significantly in a good solvent (low χ) than in a poor solvent (high χ). The interplay between the two parameters is summarized in a diagram (Fig.5) with two distinct regions for stable and unstable hydrogels. The boundary line separating the two regions is determined by setting $\mu_c^\infty = 0$ or $\lambda_c^\infty(N\nu, \chi, \phi_0) = \max(\lambda_h)$ in Eq.(3.79). The range of Nv in Fig. 6 roughly corresponds to a range between 1 kPa and 10 MPa for the initial shear modulus ($Nk_B T$) of the polymer network at 25 °C, which is typical for hydrogels and elastomers.

For a hydrogel layer with properties in the upper-right region of the diagram (stiff network, poor solvent), it swells homogeneously and remains stable at the equilibrium chemical potential ($\mu = 0$). The homogeneous swelling ratio is typically small ($\lambda_h < 3$) in this region. For a hydrogel layer with properties in the lower-left region (soft network, good solvent), surface instability occurs at a critical swelling ratio (Fig.3) before it reaches the maximum homogeneous swelling ratio. The predicted critical swelling ratio, ranging between 2.5 and 3.4, depends on both $N\nu$ and χ.

We note that, when $N\nu$ is relatively small ($< 10^{-4}$), the critical swelling ratio (Fig.3) is nearly a constant (~ 3.4) independent of Nv or χ, and the critical value of χ that separates the unstable and stable regions in Fig.5 is nearly independent of $N\nu$. On the other hand, the critical chemical potential (Fig.4), nearly independent of $N\nu$, increases with increasing χ. These results may be understood intuitively by considering the limiting case when the contribution of elasticity is negligible ($N\nu \to 0$) for both the homogeneous swelling and the stability analysis. In this case, the competition between the entropy of mixing and the enthalpy of solvent–polymer interaction dominates the swelling process. Consequently, the stability of the hydrogel layer depends on χ only. As $N\nu \to 0$, $\xi_h \to \infty$, and $\beta \to 1$. Solving Eq.(3.79) gives a nontrivial solution for the constant critical swelling ratio, $\lambda_c = 3.38$. By Eq.(2.6), the critical chemical potential is approximately

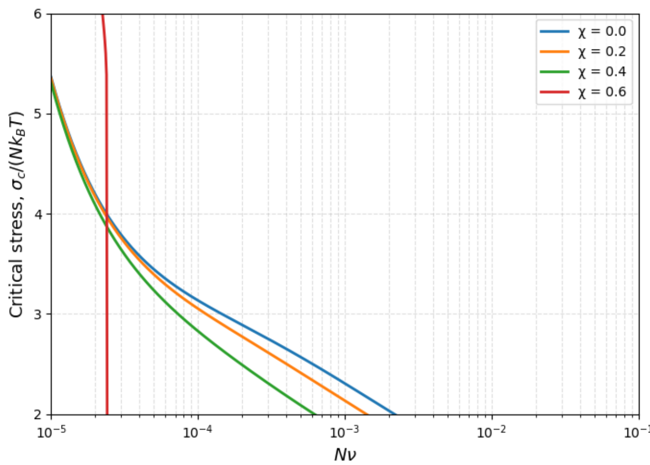

Figure 6: Critical compressive stress

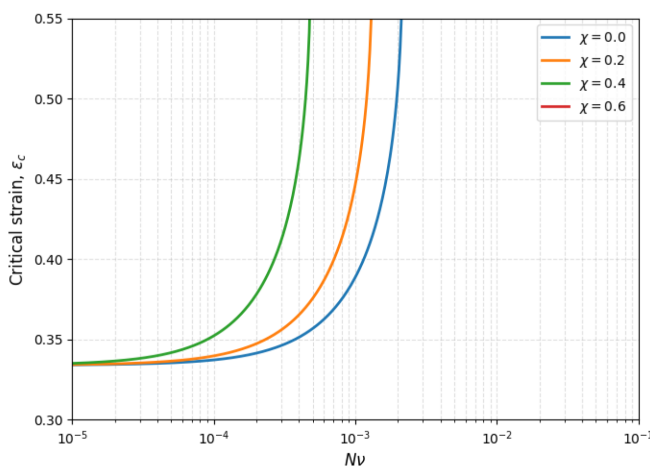

Figure 7: Critical strain

Solving equation Eq.(3.79) gives a nontrivial solution for the constant critical swelling ratio, $\lambda_c = 2.33$. By Eq.(2.6), the critical chemical potential is approximately

$$\frac{\mu_c}{K_B T} \approx 0.007368\chi - 0.003909. \tag{3.80}$$

Setting $\mu_c = 0$ in Eq.(3.80), we obtain that $\chi_c = 0.53$.

Based on the homogeneous solution in previous section, the critical compressive stress is obtained as

$$\sigma_c = NK_BT\left(\lambda_c - \frac{1}{\lambda_c}\right) + p_0 e^{\frac{\mu_c}{K_BT}}, \tag{3.81}$$

i.e.,

$$\frac{\sigma_c}{NK_BT} = \left(\lambda_c - \frac{1}{\lambda_c}\right) + \frac{p_0}{NK_BT} e^{\frac{\mu_c}{K_BT}},$$

Using the critical swelling ratio and the critical chemical potential in Fig.3 and Fig.4, we plot the critical compressive stress as a function of $N\nu$ in Fig.6.

Again, the critical compressive stress in general depends on both $N\nu$ and χ. When $N\nu < 10^{-4}$, we have approximately

$$\sigma_c \approx NK_BT\left[1.9 + \frac{\overline{p}_0}{N\nu}\exp(0.007368\chi - 0.003909)\right] \tag{3.82}$$

As given in the previous studies, the linear swelling ratio for free swelling can be obtained as a function of the chemical potential by solving the following equation:

$$\ln\left(1 - \frac{\phi_0}{\lambda_0^3}\right) + \frac{\phi_0}{\lambda_0^3} + \frac{\phi_0^2\chi}{\lambda_0^6} + N\nu\left(\frac{1}{\lambda_0} - \frac{1}{\lambda_0^3}\right) = \frac{\hat{\mu} - p\nu}{K_BT}. \tag{3.83}$$

In the other hand, the effective linear strain from the state of free swelling to that of the laterally confined swelling is then

$$\varepsilon = \frac{\lambda_0 - 1}{\lambda_0}. \tag{3.84}$$

4 Liquid phase (Passive Chemical Potential)

In [9] Eq.(3.5) we find that the pressure of the external solvent pressure is also part of the elastic part of free energy:

$$U_m(C) := \frac{K_BT}{\nu}H(J) = \frac{K_BT}{\nu}\left((J - \phi_0)\ln(1-\phi) + \chi\phi_0(1-\phi)\right), \text{with } \phi = \frac{\phi_0}{J}, \tag{4.1}$$

$$U_e(F) := \frac{G}{2}\left(|F|^2 - 3 - 2\ln J\right) + P(J-1). \tag{4.2}$$

$$W = U_e(F) + U_m(C) = \frac{G}{2}\left(|F|^2 - 3 - 2\ln J\right) + P(J-1) + U_m(C)$$

If the gel swells, the energy incrasing. This term shows that the gel invests to "fight" against the pressure of the external solvent.

Remark 4.1. *When the external solvent is in liquid state, the term related with the chemical potential is irrelevant.*

Equilibrium swelling corresponds to the case in which internal efforts cancel each other out and the stress tensor is zero. The corresponding equations are:

$$\frac{\partial U_e}{\partial F} + \frac{1}{\nu}\left(\frac{\partial U_m}{\partial C} - \mu\right) \operatorname{Cof} F = 0. \tag{4.3}$$

Let's postpone the justification for a moment. The Eq.(4.3) can be rewritten as follows:

$$GF - \left(\frac{G}{J} + \left(\frac{\mu}{\nu} - P\right) - \frac{1}{\nu}\frac{\partial U_m}{\partial C}\right) \operatorname{Cof} F = 0,$$

but when $P > P_0$, $\mu = \nu(P - P_0)$ then $\dfrac{\mu}{\nu} - P = P_0$ then μ has not effect, i.e.

$$GF - \left(\frac{G}{J} + P_0 - \frac{1}{\nu}\frac{\partial U_m}{\partial C}\right) \operatorname{Cof} F = 0. \tag{4.4}$$

Justification of Eq.(4.3)

Consider Eq.(4.3) where $U_e = U_e(F, C)$ is a mathematic function in which F and C are independents variables. (This equation becomes in physics function when coupling F and C according to $J = \phi_0 + \nu C$

$$F \mapsto U\left(F, \frac{J - \phi_0}{\nu}\right)$$

is the free energy of the system seen as function of F). The equations can be obtained from

$$\begin{cases} \sigma = J^{-1}\dfrac{\partial U}{\partial F}F^t - \Pi I, \\ \mu = \dfrac{\partial U}{\partial C} + \Pi\nu \end{cases} \tag{4.5}$$

which are the Eqs. (19) and (20) in [8]. The swelling equilibrium corresponds to $\sigma = 0$, then

$$J^{-1}\frac{\partial U}{\partial F}F^t = \Pi I,$$

therefore

$$\frac{\partial U}{\partial F} = \Pi \operatorname{Cof} F,$$

this means that elastic forces and osmotic pressure must be balanced. But

$$\Pi = \frac{1}{\nu}\left(\mu - \frac{\partial U}{\partial C}\right),$$

then the equations are

$$\frac{\partial U}{\partial F} = \frac{1}{\nu}\left(\mu - \frac{\partial U}{\partial C}\right) \operatorname{Cof} F.$$

□

The Legendre transformation

Mathematically, this equations can be seen as

$$\frac{\partial \hat{W}}{\partial F}(F, \mu) = 0,$$

where

$$\begin{aligned}\hat{W}(F, \mu) &= [U(F, C) - \mu C]_{J=\phi_0+\nu C}\,, \\ &= U\left(F, \frac{J-\phi_0}{\nu}\right) - \frac{\mu}{\nu}(J - \phi_0).\end{aligned}$$

Indeed,

$$\begin{aligned}&\frac{\partial \hat{W}}{\partial F} = 0, \\ \Leftrightarrow \quad &\frac{\partial U}{\partial F} + \frac{\partial U}{\partial C}\frac{\partial\left(\frac{J-\phi_0}{\nu}\right)}{\partial F} - \frac{\mu}{\nu}\frac{\partial J}{\partial F} = 0, \\ \Leftrightarrow \quad &\frac{\partial U}{\partial F} + \frac{1}{\nu}\left(\frac{\partial U}{\partial C} - \mu\right)\frac{\partial J}{\partial F} = 0, \\ \Leftrightarrow \quad &\frac{\partial U}{\partial F} + \frac{1}{\nu}\left(\frac{\partial U}{\partial C} - \mu\right)\operatorname{Cof} F = 0.\end{aligned}$$

□

The osmotic pressure

We find it easier to understand molecular incompressibility as an idealized caricature of extreme behavior (on the limit of very low compressibility) than as a constrained minimization problem: We define

$$U_\kappa(F, C) := U(F, C) + \frac{\kappa}{2}(1 + \nu C - J)^2.$$

The thermodynamic formulation which we study in [8] show us that

$$\sigma = J^{-1}\frac{\partial U_\kappa}{\partial F}F^T = J^{-1}\frac{\partial U}{\partial F}F^t - \kappa(1 + \nu C - J)\frac{\partial J}{\partial F}J^{-1}F^T,$$

then

$$\sigma = J^{-1}\frac{\partial U}{\partial F}F^t - \kappa(1 + \nu C - J)I.$$

Finally,

$$\mu = \frac{\partial U_\kappa}{\partial C} = \frac{\partial U}{\partial C} + \kappa(1 + \nu C - J)\nu.$$

On the limit when the cost κ to pay by the compressibility tends to infinity,

$$\kappa \to \infty, \quad 1 + \nu C - J \to 0 \tag{4.6}$$

and the product $\kappa(1+\nu C - J)$ to be part of the stress tensor σ, should be bounded and to converge to an one value (which do not need to be zero nor constant in x, y, z).

The osmotic pressure $\Pi(x, y, z)$ is the limit of this product, i.e.

$$\Pi = \lim_{\kappa\to 0} \kappa(1+\nu C - J)$$

and, on the limit,

$$\sigma = J^{-1}\frac{\partial U}{\partial F}F^t - \Pi I,$$
$$\mu = \frac{\partial U}{\partial C} + \Pi\nu.$$

Formally, seen as multiplier of Lagrange, is which is obtained if we define the augmented Lagrangian

$$\hat{U}(F, C, \Pi) = U(F, C) + \Pi(1+\nu C - J)$$

and substituting U by $\hat{U}$ in the thermodynamic equations:

$$\sigma = J^{-1}\frac{\partial U}{\partial F}F^t = J^{-1}\frac{\partial U}{\partial F}F^t - \Pi\frac{\partial J}{\partial F}J^{-1}F^t,$$

then

$$\sigma = J^{-1}\frac{\partial U}{\partial F}F^t - \Pi I$$

and

$$\mu = \frac{\partial \hat{U}}{\partial C} = \frac{\partial U}{\partial C} + \Pi\nu.$$

Finally, we observe that we can rewrite the equilibrium equation $\sigma = 0$, before to pass to the limit as $\kappa \to \infty$, as

$$\frac{\partial U}{\partial F} = \kappa(1+\nu C - J)\mathrm{Cof}\, F = \frac{1}{\nu}\left(\mu - \frac{\partial U}{\partial C}\right)\mathrm{Cof}\, F.$$

Rewriting in terms of our function $H'(J)$

In our sequel

$$\begin{aligned} U_m(C) &= \frac{K_B T}{\nu}\left((J-\phi_0)\ln(1-\phi) + \chi\phi_0(1-\phi)\right) \\ &= \frac{K_B T}{\nu}\left((J-\phi_0)\ln(1-\frac{\phi_0}{J}) + \chi\phi_0\left(1-\frac{\phi_0}{J}\right)\right). \end{aligned}$$

We can define $H(J)$ as in (4.1), i.e.

$$H(J) := (J-\phi_0)\ln(1-\frac{\phi_0}{J}) + \chi\phi_0\left(1-\frac{\phi_0}{J}\right).$$

Then

$$\begin{aligned}\Pi &= \frac{1}{\nu}\left(\mu - \frac{\partial U_m}{\partial C}\right)\\ &= \frac{1}{\nu}\left(\mu - \frac{\partial U_m}{\partial J}\frac{\partial J}{\partial C}\right) = \frac{1}{\nu}\left(\mu - \frac{K_B T}{\nu}H'(J)\frac{\partial(\phi_0+\nu C)}{\partial C}\right),\\ &= \frac{1}{\nu}\left(\mu - \frac{K_B T}{\nu}H'(J)\nu\right) = \frac{\mu}{\nu} - \frac{K_B T}{\nu}H'(J).\end{aligned}$$

Finally, the swelling equillibrium equation in (4.4) can be rewritten as

$$GF - \left(\frac{G}{J} + P_0 - \frac{1}{\nu}H'(J)\right)\operatorname{Cof}F = 0. \tag{4.7}$$

Conclusion 4.1. *(i) The osmotic pressure is not* $-\frac{K_B T}{\nu}H'(J)$, *but* $\frac{\mu}{\nu} - \frac{K_B T}{\nu}H'(J)$.

(ii) The elastic part of the free energy is not

$$\frac{G}{2}\left(|F|^2 - 3 - 2\ln J\right),$$

but

$$\frac{G}{2}\left(|F|^2 - 3 - 2\ln J\right) + P(J-1).$$

(iii) The equations of equilibrium swelling are

$$\frac{\partial U_e}{\partial F} = \Pi(Cof F),$$

with

$$\Pi = \frac{\mu}{\nu} - \frac{K_B T}{\nu}H'(J).$$

5 Finite element simulations

5.1 Ideal Gas phase (Active Chemical Potential)

If $\lambda_c > \lambda_{\text{hom}}$ we can able to find conditions such that $\lambda_c = \lambda_{\text{hom}}$. In this case we have Chemical Potential μ active.

In this case according with (1.3) the equilibrium swelling equation becomes into

$$GF - \left(\frac{G}{J} + \Psi(P) - \frac{1}{\nu}\frac{\partial U_m}{\partial C}\right)\operatorname{Cof}F = 0, \tag{5.1}$$

where

$$\Psi(P) := P - \frac{K_B T}{\nu}\ln\left(\frac{P}{P_0}\right). \tag{5.2}$$

Definition 5.1. *We say that a gel have a stable configuration if and only if for all chemical potential* $\overline{\mu} \leq 0$ *the profile of the gel in the simulation is plane (without wrinkles).*

5.2 Methods

For the finite element simulations, we discretize the computational domain with tetra- hedral meshes and use continuous piecewise cubic finite element spaces to approximate the displacement variable. All the simulations were implemented in the open-source finite element library Netgen/NGSolve (`www.ngsolve.org`) [13]. The system of nonlinear equations arising from the finite element approximation of our nonlinear boundary value problems were solved using the damped Newton's method.

5.2.1 Case ϕ_0 small enough

In this case if ϕ_0 is small enough then we have $\lambda_c < \lambda_{\text{hom}}$, for $\overline{\mu} > 0$ we have constant homogeneous swelling ratio, for these reason we choose negative values of $\overline{\mu} \leq 0$ in which we obtain Stability in the sense of the definition 5.1.

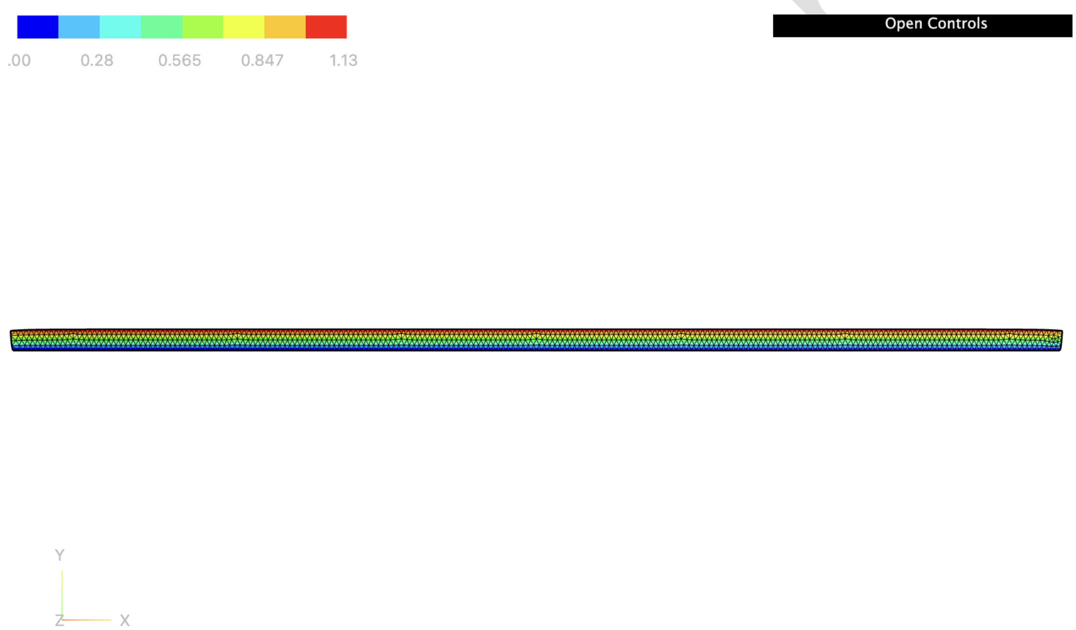

Figure 8: Successful computation of the Stable configuration for the gel for $L = 90.00$ mm, $d = 1.62$ mm, $\phi_0 = 0.20$ and $\overline{\mu} = -0.0005$ under zero-displacement Dirichlet boundary conditions ($\mathbf{u} = \mathbf{0}$) along the bonded and debonded interfaces, supplemented by a penalty-based non-penetration condition ($y + u_y \geq 0$) at the bottom surface. The nonlinear problem is solved via an incremental softening scheme across 16 load steps. Newton's method is executed with a convergence tolerance of tol $= 10^{-3}$ (and tol $= 10^{-6}$ for the final step), allowing a maximum of 100 Newton iterations per step and up to 500 iterations for the final stage The color bar is adjusted to show in red the magnitude of the maximum displacement, which is of 1.13 mm. This corresponds to the solution obtained when using the swelling equilibrium of the gel with a shear modulus of 0.1372 MPa as the initial approximation for a gel with shear modulus of 0.13721349978333722 MPa.

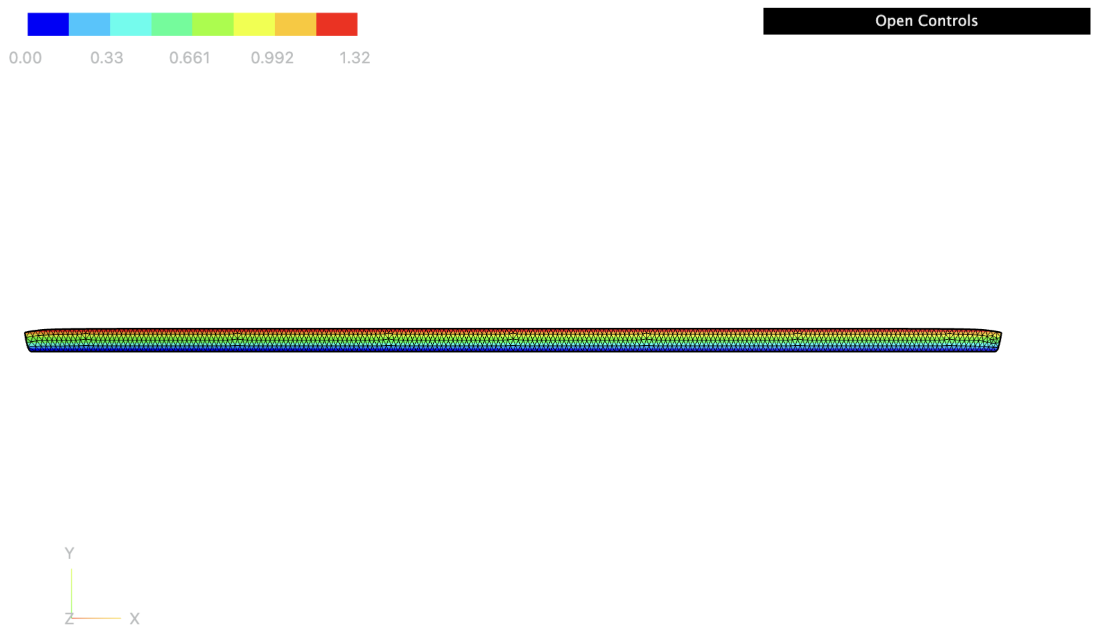

Figure 9: Successful computation of the Stable configuration for the gel for $L = 90.00$ mm, $d = 1.62$ mm, $\phi_0 = 0.20$ and $\overline{\mu} = -0.003$ under zero-displacement Dirichlet boundary conditions ($\mathbf{u} = \mathbf{0}$) along the bonded and debonded interfaces, supplemented by a penalty-based non-penetration condition ($y + u_y \geq 0$) at the bottom surface. The nonlinear problem is solved via an incremental softening scheme across 16 load steps. Newton's method is executed with a convergence tolerance of tol $= 10^{-3}$ (and tol $= 10^{-6}$ for the final step), allowing a maximum of 100 Newton iterations per step and up to 500 iterations for the final stage The color bar is adjusted to show in red the magnitude of the maximum displacement, which is of 1.32 mm. This corresponds to the solution obtained when using the swelling equilibrium of the gel with a shear modulus of 0.1372 MPa as the initial approximation for a gel with shear modulus of 0.13721349978333722 MPa.

5.3 Case ϕ_0 is larger

If ϕ_0 is larger then $\lambda_c > \lambda_{\text{hom}}$ and this suggest instability and we can refer to [9] for the case when $\phi_0 = 1$.

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